

Learning-based Denial-of-Key-Generation (DoKG) Attack Against Quantum Key Distribution

Shradha Awasthi*, Shafkat Islam*, Md Tamjid Hossain†

*Purdue University Northwest, Hammond, IN, USA

† Texas A&M University-San Antonio, USA

E-mail: {awasth10, islam59}@pnw.edu; mhossain@tamusa.edu

Abstract—Quantum Key Distribution (QKD) using the BB84 protocol enables secure communication based on quantum mechanics. While most security research addresses eavesdropping and key extraction attacks, denial-of-key generation (DoKG) attacks targeting the availability of key establishment remain under-explored. This paper presents a deep Q-network (DQN) based attacker that learns adaptive disruption strategies by exploiting classical-channel feedback in dynamic BB84 protocol. The attacker observes a six-dimensional state space derived from observable protocol parameters and selects from eleven discrete attack actions, including noisy gates, Pauli-X flips, attenuation, and strategic inactivity. The environment includes Markovian channel dynamics, adaptive photon intensity control by legitimate parties, and dynamically adjusted QBER thresholds. We evaluate the DQN approach against random and multi-armed bandit baselines across varying photon budgets and channel conditions. In the simulated BB84 environment, the DQN attacker achieved approximately 99% DoKG success while disrupting approximately 50.0–54.8% of photons, compared with 66.7–69.5% success and 58.1–59.1% photon disruption for the random baseline.

Keywords: Quantum Key Distribution, Denial-of-Key-Generation, BB84, Deep-Q-Network

I. INTRODUCTION

Next-generation wireless edge applications, including unmanned aerial vehicles (UAVs) [1], healthcare systems [2], military infrastructure [3], and satellite communications [4], increasingly depend on secure and low-latency communication for autonomous operation, real-time coordination, and mission-critical control. As quantum computing advances, the need for quantum-resistant communication mechanisms becomes more pressing. Quantum communication [5], [6] offers a promising foundation for secure key establishment by relying on fundamental properties of quantum mechanics, including the no-cloning principle and the detectability of measurement-induced disturbance. While these properties help expose eavesdropping, they do not eliminate availability risks. Disruption of the key-establishment process can prevent communication and may have serious consequences for critical systems.

Quantum Key Distribution (QKD) enables two parties to establish a shared secret key using quantum-mechanical principles. In the BB84 protocol [6]–[8], photons are transmitted over a quantum channel and measured according to randomly selected bases. Disturbance in the quantum transmission can increase the quantum bit error rate (QBER). If the estimated

QBER exceeds an acceptable threshold, the session is aborted. In this way, BB84 allows sender and receiver to detect abnormal interference during key generation and avoid using a potentially compromised key.

Much of the existing QKD security literature focuses on attacks that seek to recover or infer the secret key, including intercept-resend attacks [9], photon-number-splitting attacks [10], and detector-side attacks [11]. These studies are essential for understanding secrecy vulnerabilities and strengthening confidentiality in QKD systems. However, availability is also a core security requirement. If sender and receiver cannot generate a usable key, secure communication cannot proceed. From this perspective, preventing key generation may be operationally damaging in systems that depend on continuous secure key establishment. Prior work has recognized denial-of-service (DoS) as a practical threat to QKD [12]; however, the adversarial dynamics of sustained, resource-aware disruption during BB84 key establishment remain underexplored.

In this paper, we introduce *Denial of Key Generation* (DoKG), a QKD-specific and specialized form of DoS attack against the BB84 key-establishment process. Unlike secrecy-focused attacks that attempt to recover or infer the secret key, DoKG targets the protocol’s ability to produce a usable key. The adversary learns from observable classical-channel feedback such as abort rate, sampled QBER announced during parameter estimation, and inferred channel noise and strategically disrupts selected protocol exchanges to degrade key-generation availability. We focus on the parameter-estimation stage, where sender and receiver reveal and compare a random subset of the sifted key to estimate the QBER. Although the revealed bits are discarded from the final secret key, this exchange is essential for deciding whether the session should proceed or abort. By selectively delaying, dropping, or corrupting these classical-channel transmissions, the adversary can trigger authentication failures, inconsistent parameter-estimation records, retransmission overhead, or protocol timeouts, thereby preventing successful key generation.

The DoKG adversary seeks to sustain session failure while minimizing the amount of injected disruption. This resource-aware objective captures both attack cost and stealthiness, since excessive interference may be easily distinguishable from natural channel instability. To enable adaptive and longer-term disruption, we consider a learning-based adversary that leverages observable protocol feedback to infer changing chan-

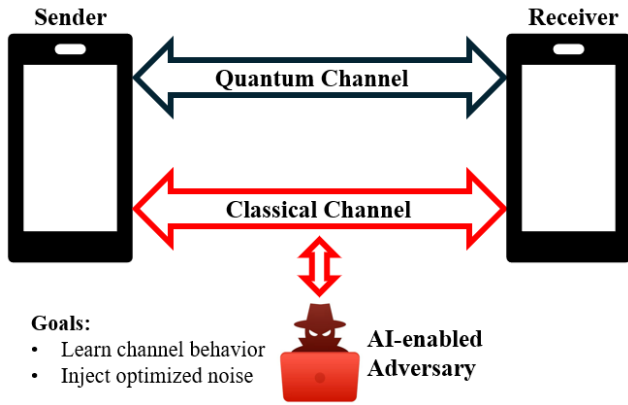


Fig. 1. Sender and receiver are establishing a quantum key for communicating through a quantum channel and an adversary is exploiting classical channel properties to inject DoKG attack during the key generation process

nel conditions and decide when and how strongly to interfere with protocol exchanges. We formulate DoKG as a sequential decision-making problem and implement a lightweight deep Q-network (DQN)-based attacker that learns disruption policies under a constrained disturbance budget. The key contributions of this paper are summarized as follows:

- 1) We present *Denial of Key Generation* (DoKG), a BB84-specific DoS attack model that targets key-establishment availability rather than key secrecy.
- 2) We formulate DoKG as an MDP and instantiate it with a lightweight DQN-based adversary that learns resource-constrained disruption policies from observable classical-channel feedback.
- 3) We conduct experiments on DoKG under dynamic channel and adaptive protocol conditions and compare the learning based DoKG attack against random baselines using availability degradation and disruption cost.

II. PROBLEM STATEMENT

Fig. 1 illustrates the adversarial QKD system with two entities, a sender and a receiver, establishing a shared secret key through BB84 protocol. A quantum channel transmits encoded photons, and an adversary observes the classical-channel feedback. The attacker observes protocol parameters to strategically inject disruptions during key establishment, forcing session aborts under a simulated bounded-disruption model. The model assumes the adversary can introduce controlled perturbations into selected protocol exchanges. This architecture captures the scenario where an adversary cannot fully control the quantum channel but can exploit the authentication and parameter-estimation stages of the protocol to achieve DoKG. We make the following assumptions regarding attacker, sender and receiver.

Adaptive sender/receiver: The sender and the receiver can adjust system parameters during operation. The mean photon number (μ) starts at μ_0 and is constrained to $[\mu_l, \mu_h]$. We assume that the abort threshold (τ) is dynamic and depends on the communication condition as $\tau = \widehat{\text{QBER}} + \alpha m_d$, but

can only fluctuate within the range of $[\tau_l, \tau_h]$. Here, $\widehat{\text{QBER}}$ is mean quantum bit error rate, m_d is mean absolute deviation of QBER and α is a balancing hyperparameter.

Attacker Observability: We assume that an attacker can only observe six features: attacker's noise estimate, abort rate, sampled quantum bit error rate, time since last abort, attacker's previous action, and inferred channel noise.

We acknowledge that the disruption actions used in this study are simulator-level abstractions and may not correspond one-to-one to immediately deployable hardware attacks against every physical QKD implementation. Rather, these actions define a bounded adversarial capability model for examining whether adaptive decision-making can increase key-generation failure under controlled BB84 simulation settings.

III. DQN-BASED DOKG ATTACK

The communication system consists of three interacting components: a dynamic communication channel, an adaptive sender and receiver, and an attacker. We formulate the DoKG attack on QKD as Markov Decision Process (MDP) based sequential decision making problem for an attacker.

A. MDP Formulation

State space: Attacker observes a six dimensional state space vector at each session. The first dimension is attacker's own noise estimate, computed as an exponential moving average of measured noise. The second dimension captures the public abort rate, representing the fraction of recent sessions that aborted. The third dimension is the sampled QBER, the QBER value announced by sender and receiver during parameter estimation. The fourth dimension tracks sessions since last abort, serving as attacker's memory of how many benign sessions have executed. The fifth dimension is the last action taken representing attacker's previous activity choice. Finally, the sixth dimension is the inferred noise which is attacker's noise estimate of the channel.

Action space: At any given state, the attacker can select an action from a discrete set of simulator-level disruption actions. An attacker can remain inactive; apply a noisy-gate disturbance at 30%, 50%, 70%, or 100% of photons, respectively; apply attenuation at 50% or 100%, respectively; apply a Pauli-X flip disturbance at 50% or 100%, respectively; combine a noisy-gate disturbance with attenuation at 50%; or combine a noisy-gate disturbance with a Pauli-X flip at 50%. These actions represent abstract disturbance classes and intensity levels in the simulator rather than device-specific attack implementations for all physical QKD systems. The action set is selected heuristically to cover representative low, moderate, high, and full disruption levels while keeping the DQN formulation tractable and comparable to the random and bandit baselines. A continuous action space could provide fine-grained control over disruption intensity and may yield a smoother cost-success tradeoff near the abort threshold; however, such a formulation would require continuous-control reinforcement learning methods and is left for future work.

Reward function: Reward incentivizes the attacker to conduct DoKG attack with minimal photon disruption.

$$r = \begin{cases} \beta_1 + \beta_2 \cdot (1 - f_{\text{attack}}) & \text{if session aborted} \\ \beta_3 + \beta_4 \cdot \frac{\text{QBER}}{\tau} & \text{if session not aborted,} \end{cases} \quad (1)$$

where f_{attack} is the fraction of photons attacked. The stealth factor $(1 - f_{\text{attack}})$ rewards attacker for achieving DoKG with fewer photons attacked. Here, $\beta_1, \beta_2, \beta_3$ and β_4 are balancing hyperparameters.

B. DQN Architecture

The DQN [13] attacker agent uses a feedforward neural network with the following architecture: an input layer accepting the 6-dimensional state vector, first hidden layer with 64 neurons and ReLU activation, second hidden layer with 32 neurons and ReLU activation, and an output layer computing 11 Q-values corresponding to each discrete action.

Experience Replay: DQN training is stabilized through experience replay, a technique that stores past transitions in a buffer, each containing (s, a, r, s') tuples (state, action, reward, next state). During training, mini-batches of 32 transitions are uniformly sampled from this buffer rather than using sequential samples from the current episode. This uniform sampling breaks temporal correlation in the data stream, preventing the network from overfitting to sequential patterns or catastrophic forgetting when channel conditions change. By repeatedly sampling uncorrelated transitions, the attacker learns robust value estimates that generalize across diverse protocol states and historical conditions, enabling effective learning without requiring massive datasets or extended training episodes.

Target Network: To provide stable learning targets, DQN maintains two neural networks, namely the policy network (updated at each training step) and a frozen target network (updated only periodically). The target network computes the Bellman target by the following equation.

$$y = r + \gamma \cdot \max_{a'} Q_{\text{target}}(s', a'), \quad (2)$$

where $\gamma = 0.95$ is the discount factor. The policy network is trained to minimize the loss $L = (Q_{\text{policy}}(s, a) - y)^2$ using stochastic gradient descent with learning rate 0.001. The target network is updated only every 50 training steps, creating a lagged copy that remains relatively stable. This decoupling prevents the network from chasing a moving target during backpropagation and allows the agent to learn from its own historical experience, significantly improving convergence stability in non-stationary QKD environments.

Epsilon-Greedy Exploration: Action selection balances exploration and exploitation through an epsilon-greedy strategy. During training, the agent selects a random action with probability ϵ and the greedy action $\arg \max_a Q(s, a)$ with probability $1 - \epsilon$. The exploration rate ϵ is initialized to 1.0 (fully random) and decayed exponentially to 0.05 by the end of training, allowing the attacker to systematically explore the action space at the beginning, then increasingly exploit learned policies as training progresses. During evaluation, $\epsilon = 0$,

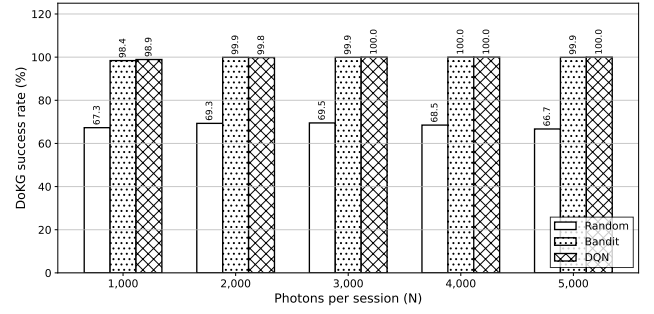


Fig. 2. Comparison of DoKG success rates across different attacker strategies

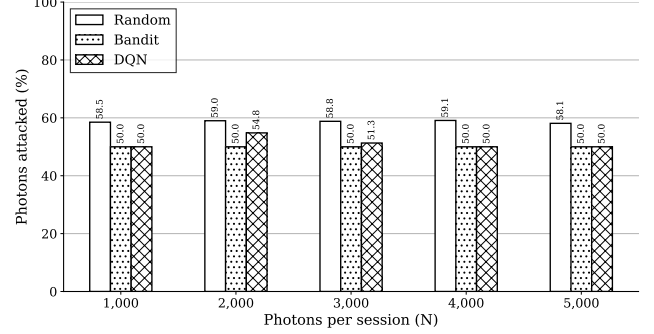


Fig. 3. Percentage of photons attacked across different attack strategies

meaning the agent acts greedily, selecting only actions with maximum Q-values.

IV. EXPERIMENTAL SETUP

We conduct the simulations on a single desktop machine equipped with an Intel Core i7-8700 processor (6 cores, 3.2 GHz), 40 GB of DDR4 RAM, and Ubuntu 22.04 LTS operating system. The implementation was developed in Python.

A. Baseline methods

To evaluate the efficacy of proposed state-aware DQN attack, we evaluate performance against two baseline strategies.

Random Strategy: The random baseline represents the most naive attack approach, wherein the adversary lacks any learning mechanism or adaptive intelligence. At each session, the random strategy uniformly samples an action from the complete set of eleven discrete actions without condition on environmental state or prior session outcomes.

Multi-Armed Bandit Strategy: The multi-armed bandit (MAB) approach represents an intermediate level of learning sophistication. The bandit strategy employs the epsilon-greedy algorithm to iteratively discover and exploit high-performing actions while minimizing action-selection variance.

B. Results

Fig. 2 shows the Denial-of-Key-Generation (DoKG) success rate with respect to photons per session (N) for three different attack strategies: random, bandit, and DQN. As photons per session (N) increases from 1,000 to 5,000, the random

strategy maintains a relatively stable but moderate success rate, fluctuating between 66.7% and 69.5%, indicating limited adaptability to changing channel conditions. The bandit approach performs significantly better, consistently achieving near-saturation performance between 98.4% and 100%, reflecting its ability to exploit reward feedback for improved attack selection. However, the DQN-based strategy achieves near-perfect performance, with success rates ranging from 98.9% to 100%, suggesting that state-aware learning can improve disruption efficiency under the simulator assumptions. Although the marginal gain of DQN over the bandit baseline is modest in absolute terms, the DQN strategy consistently maintains near-saturated DoKG success across all photon-budget settings. This suggests that state-aware policy learning can provide a small but stable advantage over action-value learning alone under the simulated dynamic-channel conditions.

Fig. 3 quantifies the percentage of photons attacked under the identical experimental conditions. The random strategy consistently attacks approximately 58.1%–59.1% of photons, indicating a lack of selectivity. The bandit strategy maintains a fixed attack rate of exactly 50% across all photon levels, suggesting a constrained or balanced exploration-exploitation policy. In contrast, the DQN approach demonstrates adaptive behavior, reducing its attack rate from 54.8% at $N=2,000$ to 50.0% at higher photon counts. This implies that the DQN agent learns to achieve near-perfect DoKG success while reducing disruption relative to the random baseline, effectively optimizing the attack efficiency. The reduction of approximately 4–9% in attack rate relative to the random baseline, while simultaneously improving success rate, highlights the effectiveness of learned policies over heuristic or static strategies.

From a scalability perspective, the results demonstrate that the DQN-based approach not only maintains but stabilizes its performance as the system scales (N increases from 1,000 to 5,000). While random strategy exhibits performance stagnation, DQN achieves near perfect DoKG success with a decreasing or stable attack budget (50 – 54.8%). Quantitatively, this translates to approximately 30–32% improvement in success rate over random and a approximately 4–9% reduction in attack overhead compared to random at scale. Importantly, the variance in DQN performance is negligible across increasing N , indicating robustness to system size and channel dynamics. This suggests that the learned policy remains stable across the simulated photon-budget settings without requiring higher disruption rates as scale increases. However, the practical feasibility of such attacks in large-scale QKD deployments requires further validation under implementation-specific physical-channel, device, and defense conditions.

V. CONCLUSION

This paper presents denial of Key Generation (DoKG), a specialized denial-of-service attack against BB84 quantum key distribution that targets availability rather than secrecy, and we formulate it as a Markov Decision Process with a

lightweight deep Q-network that learns adaptive disruption policies. Empirical results in the simulated BB84 environment show that learning based attacker can efficiently disrupt key generation in BB84 protocol compared to random baseline. However, real-world feasibility depends on implementation-specific access, device behavior, channel conditions, and defensive mechanisms. These findings reveal that availability-focused adversaries equipped with learning mechanisms can substantially disrupt key-generation with minimal resource footprint, and suggest that QKD security frameworks must address denial-of-service threats alongside conventional eavesdropping detection. Future work will develop adaptive optimization techniques for QKD in adversarial conditions, including dynamic parameter adjustment, reinforcement learning-based defenses and multi-agent scenarios to establish a comprehensive framework for mitigating availability threats to quantum communication systems.

ACKNOWLEDGMENT

We utilize AI tool (i.e., GPT 5.5) to polish the manuscript text, ensuring clarity and grammatical accuracy.

REFERENCES

- [1] Z. Gao, W. Fan, and R. Luo, "Qkd-based secure communication for uav," in *2023 24th Asia-Pacific Network Operations and Management Symposium (APNOMS)*, 2023, pp. 107–112.
- [2] S. Prajapat, D. Gautam, P. Kumar, A. K. Das, S. Pal, and C. Dong, "Blockchain-enabled secure signature scheme with quantum key distribution for iomt-based healthcare systems," *IEEE Journal of Biomedical and Health Informatics*, vol. 30, no. 4, pp. 3601–3612, April 2026.
- [3] A. Bhuyan and M. Gheorghevi, "Real-world demonstration of secure 5g roaming for military communications," in *MILCOM 2024 - 2024 IEEE Military Communications Conference (MILCOM)*, 2024, pp. 639–640.
- [4] A. Rani, X. Ai, A. Gupta, R. S. Adhikari, and R. Malaney, "Combined quantum and post-quantum security for earth-satellite channels," in *2025 International Conference on Quantum Communications, Networking, and Computing (QCNC)*, 2025, pp. 301–308.
- [5] Z. Yang, M. Zolnari, and R. Jain, "A survey of important issues in quantum computing and communications," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 1059–1094, 2023.
- [6] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," *Theoretical Computer Science*, vol. 560, pp. 7–11, 2014.
- [7] C. Lee, I. Sohn, and W. Lee, "Eavesdropping detection in bb84 quantum key distribution protocols," *IEEE Transactions on Network and Service Management*, vol. 19, no. 3, pp. 2689–2701, 2022.
- [8] B. R. La Cour and N. A. Davis, "An adversarial quantum key distribution project," in *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*. IEEE, 2025, pp. 113–118.
- [9] S. F. Hegazy, S. S. A. Obayya, and B. E. A. Saleh, "Randomized ancillary qubit overcomes detector-control and intercept-resend hacking of quantum key distribution," *Journal of Lightwave Technology*, vol. 40, no. 21, pp. 6995–7005, 2022.
- [10] M. V. Dasari, S. Kirubakaran, and A. Paventhan, "Analysis of pns attacks on quantum key distribution protocols," in *2025 IEEE Wireless Antenna and Microwave Symposium (WAMS)*, 2025, pp. 1–6.
- [11] V. Scarani and C. Kurtsiefer, "The black paper of quantum cryptography: real implementation problems," *Theoretical Computer Science*, vol. 560, pp. 27–32, 2014.
- [12] P. Schartner and S. Rass, "Quantum key distribution and denial-of-service: Using strengthened classical cryptography as a fallback option," in *2010 International Computer Symposium (ICS2010)*, 2010, pp. 131–136.
- [13] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. Riedmiller, "Playing atari with deep reinforcement learning," *arXiv preprint arXiv:1312.5602*, 2013.